**F79SU Coursework: Assignment 2**

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# Introduction

This report investigates the graduation of mortality rates for ages 35 to 80 using the data for the year 1955. The objective is to assess how well the Gompertz law describes the pattern of mortality over this age range and to evaluate alternative specifications of the model using statistical diagnostics.

As an initial exploratory step, the observed mortality rates were plotted on a logarithmic scale against age. This transformation allows the approximately exponential increase in mortality with age to be visualised more clearly and provides an indication of whether a Gompertz-type relationship is appropriate. The resulting plot suggested that the mortality pattern broadly follows a linear trend on the log scale, supporting the use of Gompertz-based models for graduation.

Three generalized linear models were then fitted under the assumption of Poisson errors, using the death counts with exposures as offsets. Each model represented a different formulation of the Gompertz relationship between age and mortality. After estimating the parameters of the models, the corresponding fitted Gompertz curves were superimposed on the initial logarithmic mortality plot to allow visual comparison between the observed data and the fitted graduations.

To evaluate the adequacy of each model, Pearson residuals were calculated and plotted against age. These plots provide a diagnostic tool for identifying systematic deviations, patterns, or serial dependence in the residuals that would indicate a lack of fit. Visual inspection of the residuals helped assess whether the models captured the main structure of the mortality pattern across ages.

A series of formal goodness-of-fit tests was then applied to the Pearson residuals using the functions provided in the TestGoF.r file. The tests conducted were: the chi-squared goodness-of-fit test, the standardised deviations test with four equal areas, the sign test, the change of signs test, the runs test, and the lag-1 serial correlation test. Together, these tests examine different aspects of model adequacy, including overall fit, bias in the residuals, independence, and the presence of systematic patterns.

Finally, the models were compared using the Akaike Information Criterion (AIC), which balances model fit against model complexity. This provided an additional quantitative measure for selecting the most appropriate model among the alternatives considered.

Overall, the analysis combined graphical examination, residual diagnostics, formal statistical tests, and information criteria to evaluate the suitability of Gompertz-type models for graduating mortality rates for the year 1955 over the age range 35 to 80.

# Task 1

Using the data generated under seed number 6014, we obtained data for the year 1955. The figure below shows the Gompertz Law for males ages 35 to 80 in that year for the CMI data set. The crude mortality rates at age $x$, $\hat{\mu}_x$, were computed by dividing the variables `DTH` with `EXP` in R. These variables contain the observed number of deaths at age $x$ and the corresponding central exposed to risk respectively.

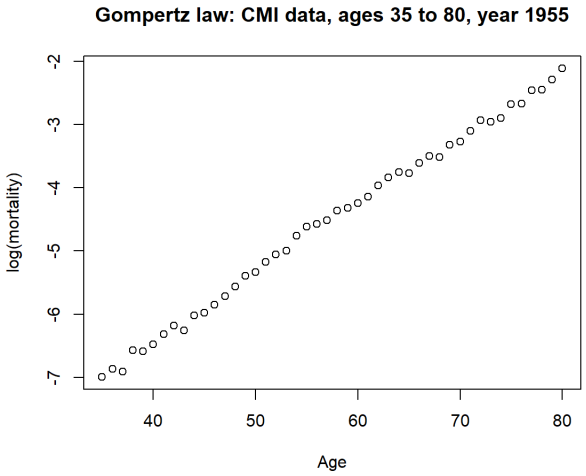



Figure 1: Log mortality for males in 1955 (CMI data).

The Gompertz Law states that log $(\hat{\mu}_x)$ is linear with age. Evidently, it is clearly depicted in the linear figure above. Hence, this linear trend supports the use of Gompertz-based models for graduation. We therefore consider this in the next task.

## Task 2

Now, assuming Poisson errors for the number of deaths at each age $x$, Generalised Linear Models (GLM)s with linear predictors of the form:

$$
\begin{aligned}
log(\mu_x) &= a_0 + a_1 x \quad \text{(Model M1)}, \\
log(\mu_x) &= b_0 + b_1 x + b_2 (x - \overline{x})^2 \quad \text{(Model M2)}, \\
log(\mu_x) &= c_0 + c_1 x + c_2 (x - \overline{x})^2 + c_3 (x - \overline{x})^3 \quad \text{(Model M3)},
\end{aligned}
$$

were fitted to the crude rates $\hat{\mu}_x$. Note that $\overline{x}$ denotes the average age in the range considered. It was computed to be 57.5.

Therefore, the fitted models are as such:

$$
\begin{aligned}
log(\mu_x) &= -10.6 + 0.1059 x \quad \text{(Model M1)}, \\
log(\mu_x) &= -10.67 + 0.1078 x - 0.0003872 (x - \overline{x})^2 \quad \text{(Model M2)}, \\
log(\mu_x) &= -10.56 + 0.106 x - 0.0004197 (x - \overline{x})^2 + 0.000007251 (x - \overline{x})^3 \quad \text{(Model M3)},
\end{aligned}
$$

## Task 3

The fitted Gompertz line (M1) was added to the initial plot in Figure 1 as such,

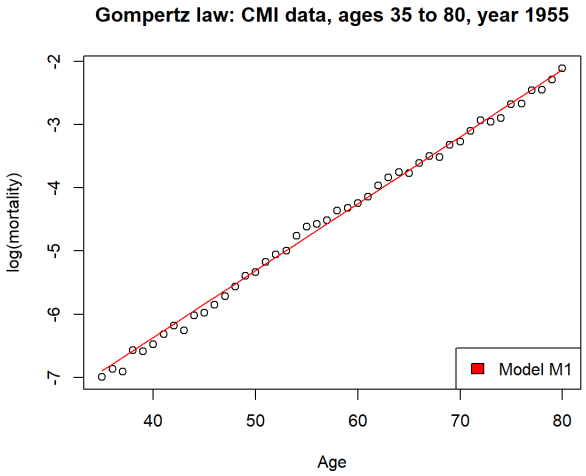



Figure 2: Log mortality for males in 1955 (CMI data), with Gompertz line (M1).

From the plot, Model M1 seems to be a good fit as it captures most of the variance in the data. Visually, the fit also follows the linear trend of the overall dataset, which indicates a good fit of the model for the data. However, it is worth noting that the model tends to very slightly overstate the mortality rates in the earlier ages before age 50. It also tends to slightly understate the mortality rates in the age ranges of 50 to around 65, followed by another slight overstatement of mortality past age 65. This is, however, as expected during graduation or smoothing of the dataset into a model. We proceed by comparing this fit with the other 2 model fit results in the next task.

## Task 4

The fitted Gompertz lines (M2) and (M3) were added to the plot in Figure 2 as such:

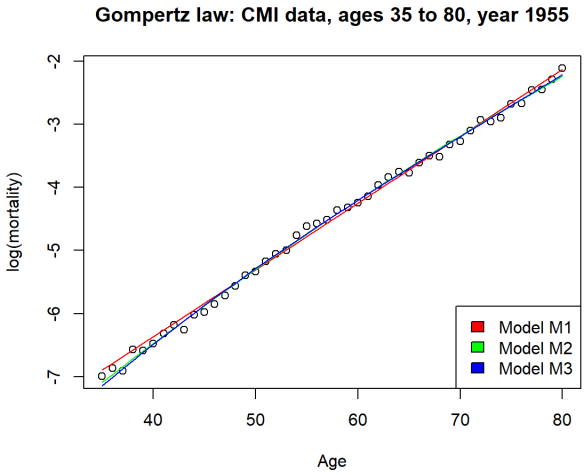


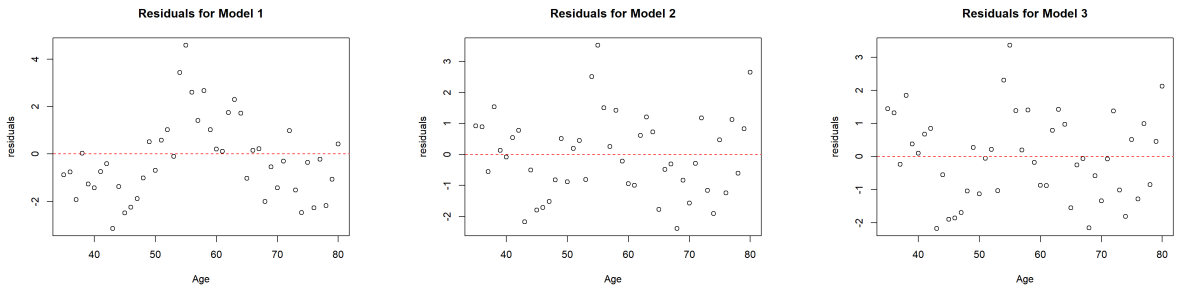

Figure 3: Log mortality for males in 1955 (CMI data), with Gompertz lines (M1), (M2) and (M3).

Models M2 and M3 seem to fit differently from model M1 by indicating lower mortality rates in the earlier ages. They also tend to indicate slightly higher mortality rates for the ages in the middle range, and fall back to slightly lower mortality rates nearing the older ages. Comparing just between models M2 and M3, there visually doesn't seem to have that much of a difference, with the only notable difference being that model M2 has slightly higher mortality rates in the younger ages.

Overall from the plot, all 3 models seems to be a good fit as it captures most of the variance in the data. Therefore, it is inconclusive as to which model is performing better solely by comparing the fit of their respective Gompertz lines on the original plot. Hence, we move on to analyse their Pearson residuals in the next segment to conduct appropriate model selection.

## Task 5

The Pearson residuals for each model were calculated (See Appendix 1) and plotted in the Figure below. Note that the Pearson Residuals were directly extracted from the model using base-R functions without any further calculations.




(a) Model M1 (b) Model M2 (c) Model M3

Figure 4: Pearson residuals in Poisson Gompertz model.

Overall, for all 3 models, there are no noticeably large values in these residuals. However, looking at Model M1, there seems to be an overall upside down "U" pattern to the Pearson residuals. Hence, this indicates that Model M1 does not provide a reasonable model for our data.

Diverting attention to Models M2 and M3, both residual plots show no obvious patterns to these residuals, other than a slight "w" shape in both plots. Hence, a battery of tests would be conducted in the later segment to further determine which model to select as the most reasonable model for our data.

## Task 6

A series of tests will be conducted on the Pearson residuals for each of the three models. Note that all alternative hypothesis henceforth are implied to be the inverse statement of the null hypothesis, unless stated otherwise.

(a) $\chi^2$ goodness of fit test

| Models | Model M1 | Model M2 | Model M3 |
|---|---|---|---|
| $\chi^2$ test statistic | 128.3022 | 78.72942 | 76.49901 |
| Degrees of Freedom | 44 | 43 | 42 |
| $p$-value | $3.549 \times 10^{-10}$ | $7.212 \times 10^{-4}$ | $9.014 \times 10^{-4}$ |

Table 1: $\chi^2$ goodness of fit test results

We compute and find the $\chi^2$ test statistic as summarised above. We have $n - k$ degrees of freedom, with $n = 46$ and $k = 2, 3, 4$ respectively for each model, reflecting the number of estimated parameters in each model. With a one-sided right side test, we compute the respective $p$-values. There is strong evidence at a 5% significance level that the graduation is unsatisfactory in all 3 models. Thus, we reject $H_0$ which states that the observed deaths are consistent with a set of graduated deaths from the model. Therefore, we conclude that the observed deaths are not consistent with a set of graduated deaths from the models. However, other tests must be done due to several weaknesses of the $\chi^2$ test including but not limited to:

- Failure to detect positive and negative bias in the residuals
- Failure to detect clumping or non-random grouping of both positive and negative signs in residuals
- Failure to detect a few large deviations offsetting an abnormally large amount of smaller deviations

It is also worth noting that $\chi^2$ tests are very sensitive with large exposures, so rejection is common even for reasonable mortality models. Hence, we move on with the remaining tests.

(b) Standard deviations test with 4 equal areas

| Models | Model M1 | Model M2 | Model M3 |
|---|---|---|---|
| Boundary points | -0.674, 0.000, 0.674 | -0.674, 0.000, 0.674 | -0.674, 0.000, 0.674 |
| Observations per cell | 21, 6, 8, 11 | 16, 8, 8, 14 | 16, 8, 8, 14 |
| Expectation | 11.5 | 11.5 | 11.5 |
| $\chi^2$ test statistic | 11.56522 | 4.434783 | 4.434783 |
| Degrees of Freedom | 3 | 3 | 3 |
| $p$-value | $9.031 \times 10^{-3}$ | 0.2181818 | 0.2181818 |

Table 2: Standard deviations test with 4 equal areas results

Details regarding the test for each model is summarised in the above table. The residuals are divided into 4 equal areas or cells, with each having an expectation of $\frac{46}{4} = 11.5$. The test statistic is the $\chi^2$ test statistic, at 3 degrees of freedom for all models. With a one-sided right side test, we compute the respective $p$-values. At a 5% significance level, we reject $H_0$ for model M1, but fail to reject $H_0$ for models M2 and M3. $H_0$ states that the residuals are normally distributed. Therefore, it can be concluded that models M2 and M3's graduation fit the data quite well.

(c) Sign test

| Models | Model M1 | Model M2 | Model M3 |
|---|---|---|---|
| Positive residuals | 19 | 22 | 22 |
| Negative residuals | 27 | 24 | 24 |
| $p$-value | 0.3019956 | 0.8829959 | 0.8829959 |

Table 3: Sign test results

Details on the number of positive and negative residuals for each model is summarised in the above table. The test statistic is the number of positive residuals, which follows a binomial distribution with parameters $n = 46$ and probability of 0.5. With a two-sided test, we compute the respective $p$-values. At a 5% significance level, we fail to reject $H_0$ which states that there is no positive or negative bias in the residuals. This conclusion holds true for all 3 models.

(d) Change of signs test

| Models | Model M1 | Model M2 | Model M3 |
|---|---|---|---|
| Sign Changes | 13 | 20 | 18 |
| $p$-value | 0.003304411 | 0.2757422 | 0.1163466 |

Table 4: Change of signs test results

Details on the number of sign changes in the residuals for each model is summarised in the above table. The test statistic is the number of sign changes, which follows a binomial distribution with parameters $n - 1 = 45$ and probability of 0.5. With a one-sided left side test, we compute the respective $p$-values. At a 5% significance level, we reject $H_0$ for model M1 but fail to reject $H_0$ for models M2 and M3. $H_0$ states that the residuals follow a standard normal distribution. Therefore, there is nothing making us suspect non-randomness in models M2 and M3.

(e) Runs test

| Models | Model M1 | Model M2 | Model M3 |
|---|---|---|---|
| Positive residuals, $n_1$ | 19 | 22 | 22 |
| Negative residuals $n_2$ | 27 | 24 | 24 |
| Groups of Positive signs | 7 | 11 | 10 |
| $p$-value | 0.006144647 | 0.3934539 | 0.1941988 |

Table 5: Runs test results

Details of the test are summarised in the above table. The test statistic is the number of groups of positive residuals, $G$, with each discrete probability being calculated as such:

$$P(G = t) = \frac{\binom{n_1-1}{t-1}\binom{n_2+1}{t}}{\binom{n_1+n_2}{n_1}}$$

With a one-sided left side test, we compute the respective $p$-values. At a 5% significance level, we reject $H_0$ for model M1 but fail to reject $H_0$ for models M2 and M3. $H_0$ states that the $n_1$ positive residuals and the $n_2$ negative residuals are in random order. Therefore, it can be concluded that the positive and negative residuals for models M2 and M3 are in random order.

(f) Lag-1 serial correlations test

| Models | Model M1 | Model M2 | Model M3 |
|---|---|---|---|
| Serial correlation | 0.5987825 | 0.3125721 | 0.2999287 |
| $p$-value | $2.950 \times 10^{-5}$ | 0.01800575 | 0.02211088 |

Table 6: Lag-1 serial correlations test results

Details of the serial correlation are summarised in the above table. The test statistic is the serial correlation which follows a normal distribution with mean 0 and variance $\frac{1}{46-1}$, due to there being $n = 46$ data and the test being lagged by 1. With a one-sided right side test, we compute the respective $p$-values. At a 5% significance level, we reject $H_0$ for all 3 models. However, it is worth noting that at a 1% significance level, we only reject $H_0$ for model M1 while failing to reject $H_0$ for models M2 and M3. $H_0$ states that the residuals are not serially correlated. Hence, models M2 and M3 show much lower serial correlation, suggesting the curvature terms improved the fit. While at a 1% significance level, we fail to reject $H_0$ for models M2 and M3, it should be noted that the serial correlation coefficient test is probably picking up the tendency for the deviations at younger ages to be positive and at the older ages to be negative. This was concluded by referring to the Pearson residuals plots done in Task 5.

## Task 7

The Akaike Information Criterion (AIC) (See Appendix 2) for each of the fitted GLM Models are summarised in the table below. Note that the AIC values here were directly extracted from the model using base-R functions without any further calculations.

| Models | Model M1 | Model M2 | Model M3 |
|---|---|---|---|
| AIC | 488.9332 | 440.8338 | 440.5588 |

Table 7: AIC results

Purely considering only the AIC values, model M3 would be the best model. This is because it has the lowest AIC value out of all 3 models. However, it is worth noting that the AIC value for Model M2 is not that much higher from the AIC value for Model M3. Hence, it is imperative to utilise all previous analysis to determine the best model to model our data.

## Conclusion (Task 8)

Based on the graphical analysis, residual diagnostics, goodness-of-fit tests and AIC comparison, Model M1 appears to be the least suitable of the three models. Although the fitted Gompertz line visually follows the overall trend in the log-mortality plot, the Pearson residuals display a clear systematic pattern, suggesting that the simple linear Gompertz specification does not adequately capture the curvature present in the mortality data. This is supported by several of the goodness-of-fit tests, where Model M1 fails a multitude of tests at the 5% significance level. In addition, Model M1 has a substantially higher AIC value compared with the other models, indicating a poorer balance between goodness of fit and model complexity.

Models M2 and M3 both provide a clear improvement over Model M1. Their Pearson residual plots show no strong systematic patterns, and the majority of the goodness-of-fit tests fail to

reject the null hypotheses, suggesting that the residuals behave approximately as expected under an adequate model. Both models also perform considerably better in terms of AIC, indicating a substantially improved fit to the mortality data.

Between Models M2 and M3, the statistical results are very similar. Model M3 has the smallest AIC value, but the improvement over Model M2 is very small. Since Model M3 introduces an additional cubic term, it is slightly more complex and may risk overfitting without providing a meaningful improvement in fit. Model M2, on the other hand, captures the curvature in the mortality pattern through a quadratic term while remaining more parsimonious.

Therefore, if one model had to be selected for applications in insurance and pensions, Model M2 would generally be preferred. It provides a good fit to the data, passes the majority of diagnostic tests, and achieves a strong balance between goodness of fit and model simplicity. This parsimony is desirable in actuarial applications, where stable and interpretable mortality models are important for reliable pricing, reserving, and long-term projections.

# A. Appendix

## Appendix 1

Formula to calculate Pearson Residuals are as follows:

$$r_i^P = \frac{y_i - \hat{\mu}_i}{\sqrt{var(\hat{\mu}_i)}}$$

where $y_i$ are the observed values, and $\hat{\mu}_i$ are the corresponding fitted values.

## Appendix 2

Formula to calculate Akaike Information Criterion (AIC) is as follows:

$$AIC = 2k - 2ln(\hat{L})$$

where $\hat{L}$ is the maximised value of the likelihood function for the model and $k$ is the number of estimated parameters in the model.